OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

Semester: V

Lab Manual

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Exercise No. 4. Simulation of Non pre-emptive CPU Scheduling algorithms

Write programs to demonstrate pre-emptive scheduling algorithms and compare their performances

Objectives

Simulation of non pre-emptive CPU Scheduling algorithms with I/O and CPU burst

times **Prerequisite**

Knowledge of scheduling algorithms

Procedure

❑ Input the number of processes to be scheduled

❑ Input the arrival time and CPU burst time of each process

❑ Calculate the turnaround time and the waiting time of the processes based on

the following scheduling methods:

- First Come First Serve (FCFS), Shortest Job First (SJF),

❑ Compare the mean turnaround time and choose the algorithm providing the

best result.

Sample Input

Enter number of processes: 3

P1 Arrival Time: 0
P1 Burst Time: 5

P2 Arrival Time: 1
P2 Burst Time: 3

P3 Arrival Time: 2
P3 Burst Time: 2

Output :(Gant chart to be displayed)

```
Enter number of processes: 3

Enter Arrival Time and CPU Burst Time:
P1 Arrival Time: 0
P1 Burst Time: 5
P2 Arrival Time: 1
P2 Burst Time: 3
P3 Arrival Time: 2
P3 Burst Time: 2

========== FCFS ==========

Gantt Chart:
0     | P1 |     5     | P2 |     8     | P3 |     10

Process AT  BT  CT  TAT WT
P1   0   5   5   5   0
P2   1   3   8   7   4
P3   2   2   10  8   6

Average Turnaround Time = 6.67
Average Waiting Time = 3.33
```

```
========== SJF ==========

Gantt Chart:
0     | P1 |     5     | P3 |     7     | P2 |     10

Process AT  BT  CT  TAT WT
P1   0   5   5   5   0
P2   1   3   10  9   6
P3   2   2   7   5   3

Average Turnaround Time = 6.33
Average Waiting Time = 3.00
```

Execution Order:

FCFS:
P1 -> P2 -> P3 -> P4 -> P5

SJF:
P1 -> P5 -> P2 -> P4 -> P3

Output performance measures

FCFS:

Throughput = 5/24 = 0.208 processes/unit time
Average Response Time = 8.20
Average Turnaround Time = 13.00
Average Waiting Time = 8.20

SJF:

Throughput = 5/24 = 0.208 processes/unit time
Average Response Time = 5.60
Average Turnaround Time = 10.40
Average Waiting Time = 5.60

Average of all the above times:

FCFS:
Average Response Time = 8.20
Average Turnaround Time = 13.00
Average Waiting Time = 8.20

SJF:
Average Response Time = 5.60
Average Turnaround Time = 10.40
Average Waiting Time = 5.60

Observation

Workout a sample test case and estimate all times, Gantt chart for both FCFS and SJF scheduling policy with CPU and I/O Burst time

Process Arrival Time CPU Burst Time
P1 0 5
P2 1 3

P3 2 8
P4 3 6
P5 4 2

FCFS Scheduling:

Execution Order:
P1 -> P2 -> P3 -> P4 -> P5

Gantt Chart:
0 | P1 | 5 | P2 | 8 | P3 | 16 | P4 | 22 | P5 | 24

SJF Scheduling:

Execution Order:
P1 -> P5 -> P2 -> P4 -> P3

Gantt Chart:
0 | P1 | 5 | P5 | 7 | P2 | 10 | P4 | 16 | P3 | 24

Conclusion:

For the given test case, SJF gives lower average waiting time and average turnaround time than FCFS. Hence, SJF provides better performance for this test case